CRH-CFG: Constraint-aware Receding-Horizon Classifier-Free Guidance for Attribute-Conditioned Flow Matching

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Abstract

Classifier-free guidance (CFG) usually uses a fixed scale and cannot respond to sample-specific generation states. We ask whether a frozen conditional–unconditional field pair can instead support online adaptation without additional generator evaluations. We introduce Constraint-aware Receding-Horizon Classifier-Free Guidance (CRH-CFG), a constrained receding-horizon search over scalar CFG actions. The search proceeds in three stages. First, it affinely reuses the field pair to construct candidate velocities and closed-form clean-endpoint estimates. Second, a frozen evaluator scores these endpoints; target-margin and protected-logit-drift constraints define the feasible set, within which the algorithm selects the action closest to ordinary conditioning and the previous choice, with a fallback when none is feasible. Third, it executes the selected action for one Heun macro-step, recomputes fields at the predicted state, and searches again after correction. This training-free search adds evaluator work but no additional generator evaluations and retains the original Heun discretization. CelebA experiments show modest adherence improvement at low overhead, while preservation and conditional quality gains remain unsupported.

1. Introduction

Flow Matching learns continuous transport by regressing a velocity field along a prescribed probability path [10]. Classifier-free guidance (CFG) modifies that field by affinely combining conditional and unconditional predictions [7, 17]. Stronger guidance can improve semantic alignment at the expense of fidelity or diversity, a trade-off also observed in classifier-guided diffusion [4].

Standard CFG uses one scale for every initial noise and every solver state. That policy cannot respond when a sample already satisfies its condition or becomes difficult only late in generation. Restricted guidance intervals, condition annealing, early suppression, and online feedback all report that the useful amount of guidance varies across time and samples [5, 8, 12, 13].

The frozen field already contains the action space needed for feedback. A conditional and unconditional pair exposes one scalar direction, $g_s = v_c - v_u$, and affine recombination produces every ordinary CFG candidate. For a straight path, each candidate also gives a closed-form estimate of the clean endpoint. This lets one field pair support a bank of evaluator-scored actions. The bank remains one-dimensional, so scale selection cannot produce an effect outside the direction already present in the model.

CRH-CFG uses this geometry as a feedback policy. At selected states, it predicts a finite endpoint bank and filters the candidates by target margin and protected-attribute drift. It then favors scales close to ordinary conditional generation and to the previous decision. An early gate avoids acting on the least reliable estimates, and a finite fallback handles empty feasible sets. Within each Heun macro-step, the selected scale is fixed while the field is reevaluated at the predicted state, keeping generator work matched.

Connecting the affine CFG line to straight-path endpoint prediction yields a vectorized, training-free controller. The controller makes its target, preservation, and minimum-intervention terms explicit. A paired, test-isolated evaluation reports the small adherence point estimate and states the limits of the preservation and quality evidence.

2. Related work

Flow Matching and classifier-free guidance. Flow Matching trains continuous normalizing flows by regressing vector fields along prescribed probability paths [10]. Guided Flows applies classifier guidance and CFG to flow-based generation [17], while standard CFG defines the affine combination of conditional and unconditional predictions [7]. CRH-CFG keeps this field frozen and changes how the scalar scale is selected.

When and how much to guide. Limited-interval guidance [8], CFG-Zero* [5], and condition annealing [13] change when or how strongly a condition acts. Guidance rescaling also depends on the noise schedule [9]. Online feedback instead uses an auxiliary evaluator to select a scale at each step [12]. CRH-CFG belongs to this last group, but restricts its actions to the ordinary CFG line and obtains its look-ahead from a closed-form endpoint estimate under a straight path. Its objective also accounts for intervention size and non-target drift.

Guidance geometry and cost. Strong extrapolation can move a sample away from the learned trajectory. CFG++ introduces a manifold-motivated correction [3], Rectified-CFG++ adapts the idea to flow models [14], and manifold projection directly addresses scale sensitivity in Flow Matching [2]. Training-based approaches can remove the cost of evaluating two branches at inference [15]. Our method does not claim a manifold guarantee or alter training. It keeps the original scalar direction and reports evaluator work separately from generator NFE.

3. From field structure to feedback

Straight-path endpoint geometry. Let $s \in [0,1]$ increase from Gaussian noise x_1 to clean data x_0 along the straight path

$$x_s = (1-s)x_1 + s x_0, \qquad \hat{x}_0(v) = x_s + (1-s)v. \quad (1)$$

For the ideal bridge, $v = x_0 - x_1$ makes the second relation exact. For a learned field, the same expression is an endpoint prediction with modeling error. The conversion between paper and implementation clocks appears in Appendix A.

One affine control axis. For condition c and null condition $\varnothing$, define $v_c = v_\theta(x_s, s, c)$, $v_u = v_\theta(x_s, s, \varnothing)$, and $g_s = v_c - v_u$. Ordinary CFG satisfies

$$v_w = v_u + w(v_c - v_u) = v_c + (w-1)g_s. \quad (2)$$

With $a_s = w_s - 1$, the sampler is control-affine,

$$\frac{dY_s}{ds} = v_c(Y_s, s) + g_s(Y_s)a_s, \quad (3)$$

and Eq. (1) maps its complete action line to

$$\hat{x}_0(w) = \hat{x}_0(1) + (1-s)(w-1)g_s. \quad (4)$$

One conditional and unconditional pair defines every scalar action and its predicted endpoint. It also fixes the local reachability boundary.

Least intervention under constraints. The ordinary conditional field corresponds to $w = 1$. We favor actions near one and near the previous decision, subject to a target margin and a protected-attribute budget. After integration, the controller observes the new state and solves the finite problem again. This produces receding-horizon feedback without a multi-step generator rollout.

4. Method: constraint-aware receding-horizon CFG

At each decision, CRH-CFG performs a constrained discrete search over $\mathcal{W}$: it enumerates scalar CFG actions, scores their estimated endpoints, filters infeasible candidates, and executes the selected action for one Heun macro-step. Algorithm 1 gives the complete search-and-execute loop before the individual components are defined.

Algorithm 1 CRH-CFG constrained search with Heun integration

Require: Frozen θ, state x, condition c, candidate set $\mathcal{W}$, evaluator F_{ctrl}

$w_{\mathrm{prev}} \leftarrow 1$
for $k = 0, \ldots, 49$ **do**
  Compute (v_u, v_c) once at the current state
  if k is a controller decision **then**
    Construct and score all endpoints by Eqs. (5) and (6)
    Select $w^\star$ by Eq. (10) or Eq. (18)
  else
    $w^\star \leftarrow w_{\mathrm{prev}}$
  end if
  Predict $\tilde{x}_{s+h}$ using $w^\star$ at (x_s, s)
  Recompute (v_u, v_c) at $(\tilde{x}_{s+h}, s+h)$
  Correct to x_{s+h} using the same $w^\star$
  $w_{\mathrm{prev}} \leftarrow w^\star$
end for
return x

4.1. Candidate endpoint bank

At each decision, the candidate set is $\mathcal{W} = \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0\}$. Conditional and unconditional fields are evaluated once, after which candidate velocities and endpoints are constructed in parallel:

$$v_{w_m} = v_u + w_m(v_c - v_u), \quad w_m \in \mathcal{W}, \quad (5)$$

$$\hat{x}_0(w_m) = x_s + (1-s)v_{w_m}. \quad (6)$$

The frozen controller evaluator scores all endpoints in one batch. For target a, desired sign y_a, and protected set P_a, we use

$$M_a(w, s) = y_a F_{\mathrm{ctrl},a}(\hat{x}_0(w)), \quad (7)$$

$$D_{\mathrm{nt}}(w, s) = \frac{1}{|P_a|} \sum_{j \in P_a} \left| F_{\mathrm{ctrl},j}(\hat{x}_0(w)) - F_{\mathrm{ctrl},j}(\hat{x}_0(1)) \right|. \quad (8)$$

The second quantity measures local attribute-logit change relative to $w = 1$. It does not measure identity or perceptual similarity.

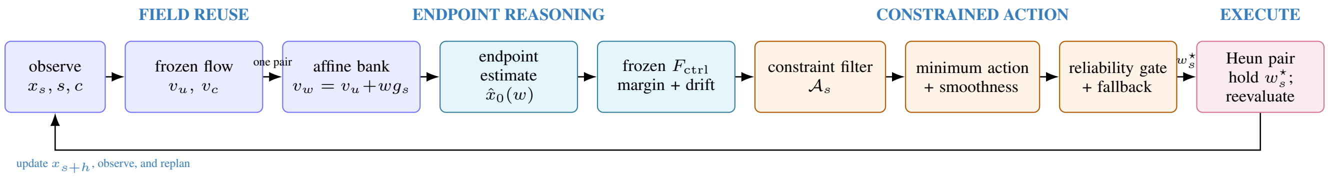

Figure 1. The CRH-CFG control loop. One frozen field pair produces the affine candidate bank and its predicted endpoints. The controller selects an action, executes it, and observes the next state. Candidate construction and endpoint scoring add no generator calls. F_{eval} remains outside the loop and measures only final outputs.

4.2. Constraint-aware minimum intervention

The feasible action set is

$$\mathcal{A}_s = \{w \in \mathcal{W} : M_a(w,s) \geq m(s), \quad D_{\mathrm{nt}}(w,s) \leq \epsilon(s)\}\,. \tag{9}$$

If $\mathcal{A}_s$ is nonempty, the controller selects

$$w_s^\star = \arg\min_{w \in \mathcal{A}_s} \rho(w-1)^2 + \eta(w - w_{\mathrm{prev}})^2. \tag{10}$$

The controller first enforces the online constraints, then minimizes movement from ordinary conditional generation and the previous action. When the feasible set is empty, a finite hinge-penalty objective returns the candidate with the lowest violation cost and records which constraints failed. Appendix B gives the schedules, normalization, and fallback.

4.3. Receding-horizon execution

Validation selected $w = 2$ during $s < \tau_{\mathrm{on}} = 0.1$, when endpoint estimates are least reliable. After the gate, the controller runs every five Heun macro-steps and holds the previous scale between decisions.

Heun-consistent scale holding. At a controller decision state x_s, CRH-CFG first selects $w_s^\star$. This scalar is held fixed over the complete Heun macro-step. The predictor evaluates $v_{w_s^\star}(x_s, s)$ and forms a provisional state $\tilde{x}_{s+h}$. The conditional and unconditional fields are then recomputed at $(\tilde{x}_{s+h}, s+h)$ and recombined using the same $w_s^\star$ for the corrector. The controller replans only after the corrected state x_{s+h} is obtained. Scale holding therefore fixes the control action within a macro-step; it does not reuse the predictor velocity.

Each Heun macro-step requires two guided-field evaluations, one at the current state and one at the predicted state. Each guided-field evaluation comprises one conditional and one unconditional U-Net forward. Across 50 macro-steps, static CFG and CRH-CFG therefore both execute 100 guided-field evaluations, or 200 raw U-Net forwards per trajectory. The candidate bank is constructed by affine recombination of the field pair already computed at a decision state. CRH-CFG adds ten batched evaluator decisions but no generator evaluations or additional generator trajectories. Appendix B gives the Heun update and complexity.

5. Experimental protocol

Frozen generator and data. The generator is an 18.5M-parameter, 64×64 attribute-conditioned U-Net with a 50,000-iteration EMA checkpoint. Its weights remain fixed. The inherited CelebA subset contains 63,715 images [11], and a deterministic split separates controller training, validation, and final test data. The test requests are positive Smiling, Bangs, and Blond Hair. Appendix C lists the architecture and split counts.

Methods and paired protocol. We compare static $w = 2$, validation-selected interval CFG, target-only feedback, target-plus-preservation feedback, and full CRH-CFG. All methods use the same checkpoint, EMA weights, conditions, initial noises, and 50-step Heun discretization. Each target has 1,000 shared seeds; the ablations use the first 256. Controller schedules and penalties are selected on validation data and then frozen. Appendix C records their values.

Evaluators and metrics. F_{ctrl} selects actions. A separately fine-tuned F_{eval} measures final target success and protected-logit drift. Both evaluators use ImageNet-pretrained ResNet-18 features [6]; separate heads and fine-tuning cannot remove all correlated representation error. CelebA annotations add another source of error [16]. Preservation is measured against paired, same-seed $w = 1$ outputs. KID [1] uses the full inherited CelebA marginal. It measures marginal closeness, not target-conditional quality. The appendix reports evaluator accuracy, KID sampling details, and metric definitions.

6. Results

6.1. Main comparison

Full CRH-CFG changes unweighted mean success from 0.9840 to 0.9880 in Table 1, a difference of $+0.4$ percentage points. The counts change from 985 to 987 for Smiling, 980 to 984 for Bangs, and 987 to 993 for Blond Hair, with 1,000 samples in each case. The stored artifact contains aggregate counts but omits the per-seed success transitions needed for McNemar or paired-bootstrap uncertainty. We

Table 1. Main results on 1,000 paired seeds per target. All methods use the same frozen generator and 50-step Heun discretization. Drift and flip are relative to the paired static $w = 2$ result; $\mathrm{KID}_{\mathrm{marg}}$ is mean $\pm$ subset standard deviation against the inherited full-marginal reference. All rows use 100 guided-field evaluations.

Target	Method	Succ. ↑	Δ pp	Drift ↓	Flip ↓	$\mathrm{KID}_{\mathrm{marg}}$ ↓	$\bar{w}$	Δ time
Smiling	Static $w = 2$	.985	+0.0	.0000	.0000	.01223±.00063	2.0000	+0.00%
	Interval	.979	-0.6	.1169	.0037	.01173±.00060	N/A	-0.50%
	Target only	.860	-12.5	1.5417	.0498	.00904±.00050	1.0739	+2.30%
	Target + pres.	.860	-12.5	1.5417	.0498	.00904±.00050	1.0739	+2.54%
	Full CRH-CFG	**.987**	**+0.2**	.0025	.0003	.01223±.00063	2.0067	+2.70%
Bangs	Static $w = 2$	.980	+0.0	.0000	.0000	.04110±.00162	2.0000	+0.00%
	Interval	.972	-0.8	.2707	.0093	.04087±.00161	N/A	-0.07%
	Target only	.888	-9.2	3.1288	.1258	.02379±.00126	1.1952	+1.61%
	Target + pres.	.888	-9.2	3.1288	.1258	.02379±.00126	1.1952	+1.97%
	Full CRH-CFG	**.984**	**+0.4**	.0476	.0018	.04123±.00162	2.0480	+1.58%
Blond	Static $w = 2$	.987	+0.0	.0000	.0000	.06872±.00225	2.0000	+0.00%
	Interval	.986	-0.1	.3156	.0095	.06711±.00221	N/A	+0.07%
	Target only	.972	-1.5	1.7170	.0555	.05884±.00206	1.2571	+1.47%
	Target + pres.	.972	-1.5	1.7170	.0555	.05884±.00206	1.2571	+1.29%
	Full CRH-CFG	**.993**	**+0.6**	.2858	.0070	.07083±.00229	2.1689	+1.12%

Table 2. Mean cosine distance between paired random samples in frozen F_{eval} backbone features. Higher denotes greater feature diversity. Target-only and target-plus-preservation are identical.

Target	Static	Interval	Target	Full
Smiling	.3729	.3750	.4383	.3724
Bangs	.4290	.4344	.4858	.4281
Blond	.2579	.2637	.2885	.2515

treat the difference as a descriptive point estimate; statistical reliability is unknown.

Relative to static $w = 2$, full CRH-CFG changes $\mathrm{KID}_{\mathrm{marg}}$ by $+0.00001$, $+0.00013$, and $+0.00211$ for Smiling, Bangs, and Blond Hair. The first two changes are small relative to the reported subset standard deviations, which are not confidence intervals for method differences. Since the reference mixes positive and negative examples, the increases cannot identify a decline in conditional quality. The experiment also finds no improvement under the common preservation reference. This KID protocol leaves conditional quality unresolved, and the intended joint benefit is unsupported.

Table 2 reports the frozen-feature diversity diagnostic. Target-only control selects much lower scales and has greater diversity for every attribute. Full CRH-CFG stays close to static CFG for Smiling and Bangs, but Blond Hair decreases from .2579 to .2515, or 2.46%. The unweighted mean changes from .3532 to .3507. This proxy has no confidence interval, so the values remain descriptive. The measured methods trade off success, marginal KID, protected drift, diversity, and wall time; none dominates every metric.

6.2. A common preservation reference

Static CFG has zero drift in Table 1 by construction, so that table cannot show whether it preserves the ordinary conditional solution. Table 3 gives every method the same paired $w = 1$ reference and retains target-wise records before averaging.

Table 3. Unweighted mean preservation diagnostic against common, same-seed $w = 1$ outputs. Lower drift and flip imply closer protected attributes, but can coincide with lower target success.

Method	Success ↑	Drift ↓	Flip ↓
Static $w = 2$	.9840	2.7770	.0963
Interval	.9790	2.7596	.0973
Target / target+pres.	.9067	1.1067	.0318
Full CRH-CFG	.9880	2.8433	.0980

Table 4. Component ablation on 256 seeds per target, reported as unweighted target means. KID uses the reduced 100-image/50-subset full-marginal-reference protocol and is not comparable with Table 1.

Variant	Succ.	Drift	Flip	KID
Static $w = 2$	.9857	.0000	.0000	.04152
Target feedback	.8503	2.9184	.1104	.02595
Target + pres.	.8503	2.9184	.1104	.02595
Target + min. int.	.9063	2.0954	.0749	.03070
Target + pres. + min.	.9063	2.0954	.0749	.03070
Full package	.9883	.0833	.0036	.04224
Full, no gate	.9154	1.8887	.0677	.03480
Full, fixed threshold	.9870	.5143	.0169	.04476

Full CRH-CFG is slightly farther from $w = 1$ than static CFG on both common reference metrics. Target-only and target-plus-preservation remain much closer, but lose 7.73 mean success points. The selected controller exposes a tradeoff between target success and preservation. It does not improve preservation.

Figure 2. Paired Blond Hair generations for the first 16 frozen seeds. Rows are static $w = 2$, interval CFG, target-only control, target-plus-preservation, and full CRH-CFG. The two reduced controllers are visually and numerically identical, while the full controller remains close to static CFG. This grid is seed-ordered and not selected by image quality.

6.3. Component and hyperparameter evidence

Feedback without the full regularized package moves toward $w = 1$ and loses adherence in Table 4. Adding the preservation term changes none of the reported values, with or without minimum intervention, because the selected budget is non-binding in these variants. Removing the early reliability gate reduces mean success from 0.9883 to 0.9154. That comparison supports the complete configuration but cannot isolate smoothness from the other changes made at the same time.

The preregistered $w_{\max}$ sweep shows little benefit from a larger action set. Increasing $w_{\max}$ from 2 to 3 changes success from 0.9857 to 0.9883, while drift rises from 0.0004 to 0.0775 and reduced KID rises from 0.04152 to 0.04215. Increasing $w_{\max}$ to 4 leaves success at 0.9883. The sweep over $\epsilon(1) \in \{0.25, 0.5, 1.0\}$, whose final decision budgets are $\epsilon(.9) \in \{.425, .650, 1.100\}$, produces identical aggregate values. The preservation constraint does not determine the selected actions in this configuration.

6.4. Controller dynamics and failures

The controller changes scale for a minority of difficult samples, while the population policy stays near static CFG. At $s = 0.9$, mean scales are 2.0170, 2.0850, and 2.1855 for Smiling, Bangs, and Blond Hair. The median and interquartile scale remain 2 at every saved decision. Final fallback rates are 0.057, 0.176, and 0.016; feasible rates are 0.943, 0.824, and 0.984. Bangs has the lowest feasible rate. Blond Hair receives the strongest late intervention and also has the largest positive adherence point estimate and the largest increase in marginal-reference KID.

The rule-selected failures make the reachability limit concrete. On Smiling seed 410577, the signed margin changes from -10.291 to -10.139, and the controller falls back on 30% of decisions despite reaching $w = 3$. On Bangs seed 410550, fallback reaches 80%; the largest selected scale is 2, and the margin falls from -6.201 to -7.777. On Blond Hair seed 410327, the controller reaches $w = 4$ and moves the margin from -7.205 to -5.250 without crossing the de-

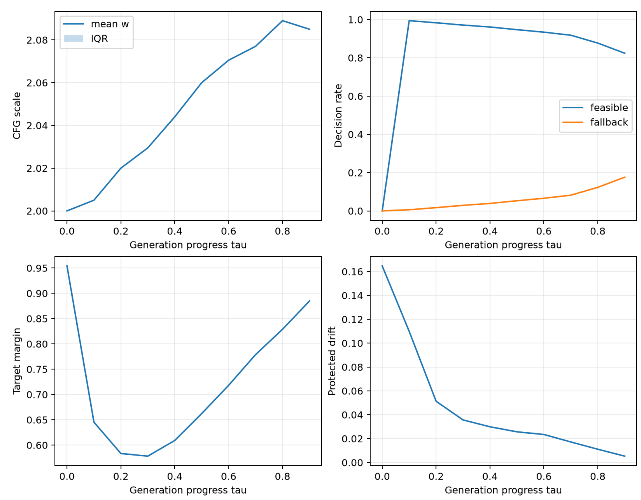

Figure 3. Full-controller dynamics for Bangs over 1,000 trajectories. The median and interquartile scale remain 2, while the fallback rate rises late in sampling. This is the target with the highest final fallback rate.

Table 5. Controller state at the final saved decision $s = 0.9$. Although mean scale differs by target, all scale medians and interquartile ranges equal 2.

Target	Mean w	Feasible	Fallback
Smiling	2.0170	.943	.057
Bangs	2.0850	.824	.176
Blond Hair	2.1855	.984	.016

(a) Smiling (b) Bangs (c) Blond Hair

Figure 4. Rule-selected paired cases. Within each panel, columns are static $w = 2$, interval CFG, target-only control, and full CRH-CFG; rows are the maximum-margin success, the case closest to the independent evaluator boundary, and the minimum-margin failure. Frozen statistics determine the selection.

cision boundary. The scalar action set cannot resolve these samples. Scores on predicted endpoints from F_{ctrl} also do not transfer perfectly to final outcomes under F_{eval}.

6.5. Cost

Full CRH-CFG uses the same generator work as static CFG and adds 1.80% mean wall time across targets. Candidate tensors and evaluator batches increase peak memory from 644,830,208 to 673,739,776 bytes, a difference of 27.6 MiB. Calibration, main experiments, ablations, and analysis consumed 3.824 L40S GPU-hours. Retained HW3 timestamps indicate another 2.6875 GPU-hours for generator training; this is a timestamp-based estimate.

The artifact audit verifies seed coverage, sample and trajectory counts, finite metrics, checkpoint hashes, the deterministic split hash, and agreement between CSV rows and paper tables. Nineteen automated tests cover the time convention, endpoint prediction, candidate vectorization, fallback behavior, Heun scale holding, paired-seed reproducibility, split isolation, checkpoint freezing, and matched generator NFE. These checks address implementation and protocol errors. They do not validate the semantic proxy.

7. Discussion and limitations

What is supported. The full policy behaves as a conservative correction around $w = 2$. Smoothness and the early gate prevent the movement toward $w = 1$ seen in reduced variants, which preserves adherence at matched generator work. The resulting adherence difference is small. Because the preservation term is non-binding, the full policy inherits the preservation behavior of static CFG. The experiment supports low-overhead adaptive correction but not the intended joint benefit.

Structural and proxy limits. Equation (3) provides one local control direction. If every candidate on that line violates a constraint, replanning cannot create a new action. Endpoint estimates are least reliable near noise, and both online constraints depend on F_{ctrl}. The final evaluator has separate weights but shares the architecture and pretrained feature source. Evaluator ensembles and independently validated control directions may address these limits, but each would require a new matched-compute baseline.

Evidence limits. Static CFG records 2,952 successes across 3,000 trajectories, while CRH-CFG records 2,964. The missing per-seed outcome transitions prevent a paired significance test. KID subset deviations are not confidence intervals, and the marginal reference cannot identify conditional quality. The evaluation covers one checkpoint, dataset, resolution, solver, and three positive requests. It does not test generalization to other models or data regimes.

Scope and responsible use. Protected-logit drift covers only the selected attributes. It does not measure identity, perceptual similarity, fairness, or consent. CelebA labels are noisy and correlated [16], and appearance labels can encode social assumptions. These experiments do not support real-person editing or reliable control of sensitive attributes.

8. Conclusion

CRH-CFG combines the affine CFG action line with straight-path endpoint prediction to provide finite, minimum-intervention feedback for a frozen flow. It retains the same 50-step Heun discretization and generator evaluation count as static CFG and adds 1.80% mean wall time. Across three targets with 1,000 paired seeds each, mean success changes by a descriptive $+0.4$ percentage points. Common-reference preservation does not improve, and marginal-reference KID cannot determine conditional quality. The method provides low-overhead adaptive correction within a one-dimensional action set. Evidence for a general joint improvement is absent.

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A. Time conventions and endpoint derivation

The paper-time convention runs from data to noise,

$$x_t = (1-t)x_0 + tx_1, \qquad t = 0 \text{ data}, \quad t = 1 \text{ noise}. \tag{11}$$

Its ideal bridge velocity is $u_t = x_1 - x_0$, so

$$x_t - tu_t = (1-t)x_0 + tx_1 - t(x_1 - x_0) = x_0. \tag{12}$$

The implementation uses the native generation clock $s = 1 - t$ and the reverse velocity $v_s = -u_t = x_0 - x_1$. Substitution gives

$$\hat{x}_0 = x_s + (1-s)v_s = x_t - tu_t. \tag{13}$$

The equality is exact for the ideal straight bridge. With a learned velocity, the expression becomes a closed-form one-step endpoint estimate.

For CFG, $v_w - v_1 = (w-1)(v_c - v_u) = (w-1)g_s$. The native endpoint map then implies

$$\begin{aligned} \hat{x}_0(w) - \hat{x}_0(1) &= (1-s)(v_w - v_1) && (14) \\ &= (1-s)(w-1)g_s, && (15) \end{aligned}$$

which yields Eq. (4). The relation explains the computational reuse and the one-dimensional reachability limit. It does not imply that evaluator scores are monotonic in w.

B. Controller and solver details

Schedules and normalization. Each controller-evaluator margin is divided by its validation-only standard deviation, which preserves the zero decision boundary. The target threshold and protected budget are

$$\begin{aligned} m(s) &= m_{\text{early}} + (m_{\text{late}} - m_{\text{early}})s^{\gamma}, && (16) \\ \epsilon(s) &= \epsilon_{\text{early}} + (\epsilon_{\text{late}} - \epsilon_{\text{early}})s. && (17) \end{aligned}$$

Decisions occur at $s \in \{0, .1, \ldots, .9\}$. The late schedule parameters specify the value at $s = 1$, which is beyond the final observed decision. Under the frozen configuration, that decision uses $m(.9) = .567$ and $\epsilon(.9) = .650$.

Fallback. When $\mathcal{A}_s$ is empty, CRH-CFG minimizes the finite objective

$$\begin{aligned} J_s(w) = \alpha\,[m(s) - M_a(w,s)]_+ + \lambda\,[D_{\text{nt}}(w,s) - \epsilon(s)]_+ \\ + \rho(w-1)^2 + \eta(w - w_{\text{prev}})^2, \qquad w \in \mathcal{W}. \end{aligned} \tag{18}$$

The finite, nonempty set $\mathcal{W}$ guarantees that the fallback returns a candidate. The implementation logs target and preservation violations separately, along with their joint occurrence.

Heun integration. With $h = 1/50$, predictor and corrector share the selected action:

$$\begin{aligned} \tilde{x}_{s+h} &= x_s + h\, v_{w_s^\star}(x_s, s), && (19) \\ x_{s+h} &= x_s + \frac{h}{2}\left[v_{w_s^\star}(x_s, s) + v_{w_s^\star}(\tilde{x}_{s+h}, s+h)\right]. && (20) \end{aligned}$$

The conditional and unconditional branches in the second guided-field term are recomputed at the predicted state. The shared $w_s^\star$ fixes the control action, not the velocity, within the macro-step. Replanning occurs only after x_{s+h} has been obtained. Algorithm 1 gives the complete inference loop in the main paper.

For batch size B and $M = |\mathcal{W}|$, the candidate endpoints form an $M \times B \times C \times H \times W$ tensor. The implementation clips the tensor to $[-1, 1]$ and evaluates it in one batch. Each decision requires $O(MBCHW)$ tensor work and one controller-evaluator pass over MB images. No additional generator trajectory is launched.

C. Experimental details

The frozen conditional U-Net has base width 64 and channel multipliers $[1, 2, 2, 4]$. Each level has two residual blocks; attention operates at 16×16 with four heads. The model contains 18,535,619 parameters. A seed-42 permutation fixes 50,972 training, 6,371 validation, and 6,372 final-test images. Main experiments use seeds 410000 through 410999 for each target and method. Ablations use the first 256 seeds.

Table 6. Validation-frozen controller configuration.

Component	Value
Candidate scales	$\{0.5, 1, 1.5, 2, 3, 4\}$
Target schedule	$m(0) = -.5$, $m(1) = .75$, $\gamma = 1.5$
Drift budget	$\epsilon(0) = 2.0$, $\epsilon(1) = .5$, linear
Penalties	$(\alpha, \lambda, \rho, \eta) = (1, 1, .05, .20)$
Reliability gate	$s < 0.1$, default $w = 2$
Decision period	Every 5 of 50 macro-steps

F_{eval} starts from the same fixed ImageNet initialization as F_{ctrl}, but uses new heads and training seed 31,415. Training uses the inherited training indices, and model selection uses validation data. The final-test labels are opened only during the final evaluation phase.

Main KID uses 1,000 generated images and the complete inherited real marginal of 63,715 images. Its subset size is 1,000, with 100 fixed subsets. Positive examples make up 44.42% of the Smiling reference, 15.87% of the Bangs reference, and 11.13% of the Blond Hair reference. Ablations use a subset size of 100 and 50 subsets. Results from the two protocols are kept separate.

Table 7. Separately fine-tuned F_{eval} performance on real final- test images. These labels are not used by the controller.

Attribute	Accuracy	AUROC
Smiling	.8991	.9626
Bangs	.9510	.9771
Black Hair	.8788	.9335
Blond Hair	.9317	.9684
Brown Hair	.7960	.8786

D. Metric definitions

Let N paired samples be indexed by i, target sign be y_a, and final independent-evaluator margin be $z_{i,a}$. Target success is

$$\text{Success}_a = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}[y_a z_{i,a} \geq 0]\,. \tag{21}$$

For reference method r and protected set P_a, paired logit drift is

$$\text{Drift}_{a,r} = \frac{1}{N|P_a|}\sum_{i=1}^{N}\sum_{j\in P_a}\left|z_{i,j}^{\text{method}} - z_{i,j}^{r}\right|. \tag{22}$$

Protected flip rate counts changes in the margin sign. For the common-reference diagnostic, r is the paired ordinary conditional trajectory with $w = 1$.

E. Reproducibility ledger

Table 8. Frozen experiment ledger.

Item	Frozen value
Generator	18,535,619-parameter conditional U-Net; EMA checkpoint; no HW4 updates
Integrator	50 Heun macro-steps; selected scale held per predictor-corrector pair; fields recomputed at the predicted state
Main seeds	410000 through 410999 for each target and method
Ablation seeds	First 256 main seeds for each target
Controller	Ten decisions per trajectory; gate for $s < 0.1$
Main KID	1,000 generated; 63,715 real; subset 1,000; 100 subsets
Reduced KID	256 generated; subset 100; 50 subsets
Tests	19 implementation and protocol tests passed

F. Failure records

Local feedback does not reverse the Smiling failure. Every late candidate for the Bangs case misses the target threshold. A stronger intervention improves the Blond Hair margin without crossing the success boundary.

Table 9. Rule-selected failures with independent-evaluator signed margins.

Target / seed	Static	Full	Max w	Fallback	Drift
Smiling / 410577	-10.291	-10.139	3	.30	.184
Bangs / 410550	-6.201	-7.777	2	.80	1.197
Blond / 410327	-7.205	-5.250	4	.10	.126